XD Technology Consulting Ltd

CUSTOMER SEGMENTATION

Application of ML Models for Clustering Customer Data

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2 Introduction

Customer segmentation is the process of grouping customers into distinct segments based on shared characteristics or behaviours. Improved understanding of customer groups enables the application of targeted marketing strategies, which can lead to increased customer satisfaction, retention, and long-term growth.

2.1 Purpose of this Report

This report details the application of unsupervised machine learning techniques and clustering models to perform customer segmentation, using data from customer purchases. The effectiveness of the resulting segmentation is evaluated, key customer characteristics for each identified group are identified with corresponding recommendations for marketing actions.

Recommendations for further development and next steps are provided in the conclusions.

3 Methodology

The approach taken involved three main stages, summarised below:

1. Data exploration and feature engineering – Includes fundamental data quality checks, and **creating a customer-level feature dataset** by aggregation of transactional e-commerce order data.
2. Applying unsupervised Machine Learning (ML) methods to **create clustering models** and using cluster number selection and validation methods to **determine the optimum number of clusters**.
3. **Segmentation of the customer base** by applying K-Means for the determined number of clusters, and applying statistical analysis to distinguish customer characteristics. Visualisation of clusters to validate the cluster segregation and highlight any overlap.

Details are provided in the following sections.

3.1 Data Summary and Verification

The dataset contains 951,669 rows, each representing a customer order line. Checks for duplicate values, missing values, and invalid data types were completed. Removing 21 exact duplicate rows left 951,648 rows for analysis. Identical rows were assumed to be duplicate observations.

3.2 Feature Engineering and Exploratory Data Analysis (EDA)

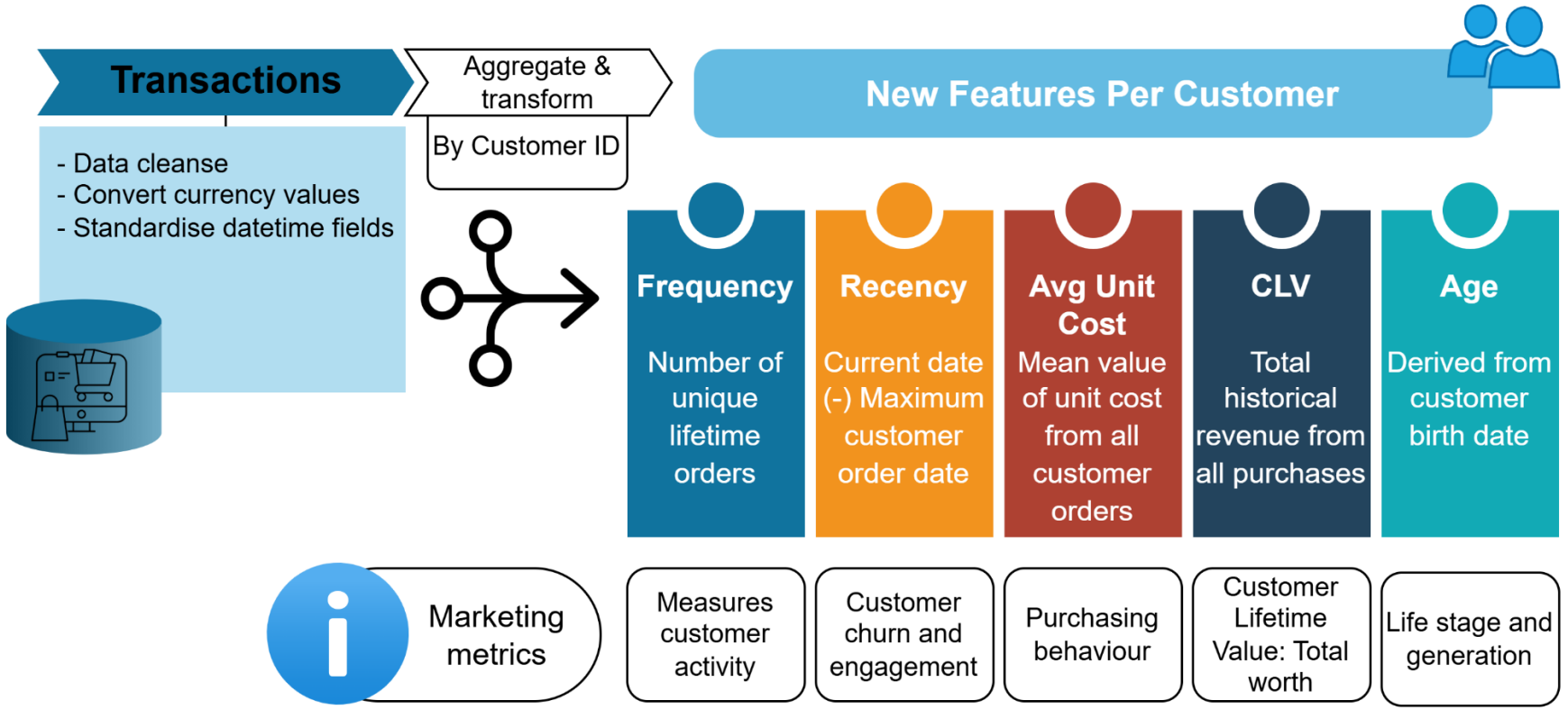


Figure 1 – Feature Engineering Process & Definitions

In this analysis, CLV refers to total revenue observed during the dataset period, rather than predicted lifetime value. Age and recency are calculated as at 26 January 2017. Average unit cost is the unweighted mean of order-line unit costs.

Descriptive statistics and visual inspections (via box-plots, histograms and Q-Q plots) were used to assess new feature distributions and identify any outliers.

3.3 Clustering Approach

(i) Determining k (the optimal number of clusters) is required for effective K-Means clustering, and this can be challenging without domain knowledge or insights. Several methods were employed as detailed below.

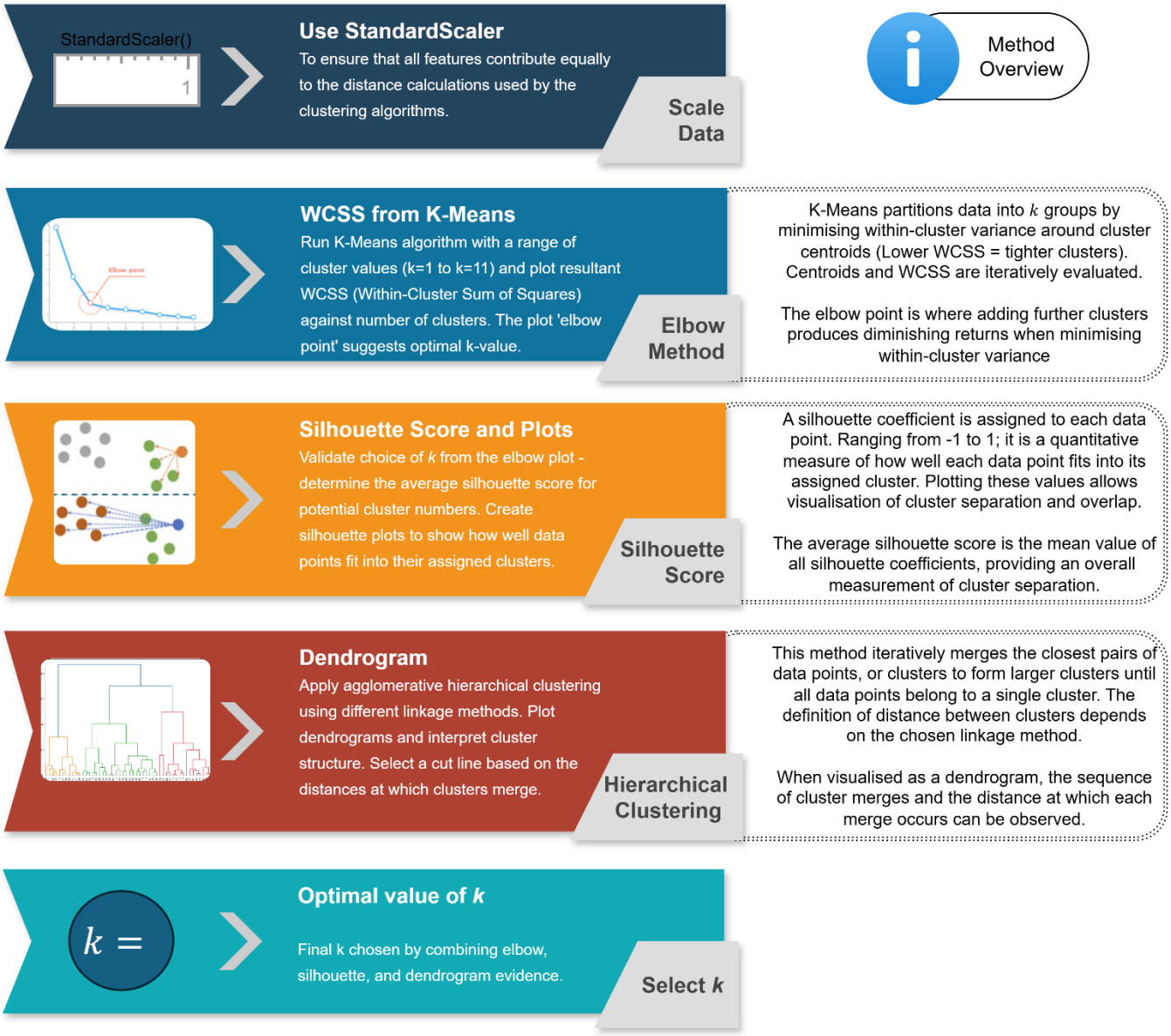


Figure 2 – Steps to determine number of clusters

(ii) Perform K-Means clustering to achieve customer segmentation. Assign cluster numbers (labels generated from the K-Means model for the chosen value of k) to the original aggregated feature labelled data. Generate per-cluster box blots for each of the features and use statistical analysis to assign distinguishing customer characteristics. Group customer groups into marketing categories and record actionable marketing insights.

(iii) Visualise the data by performing dimensionality reduction with PCA and t-SNE and plotting the clustered data coloured to match the box-plots from the statistical analysis. Determine the cluster separation achieved.

4 Results

4.1 Data Analysis & Assessment

Five customer-level features were assessed. Customer ID was retained as an identifier and excluded from modelling. Each row represents marketing metrics for an individual customer.

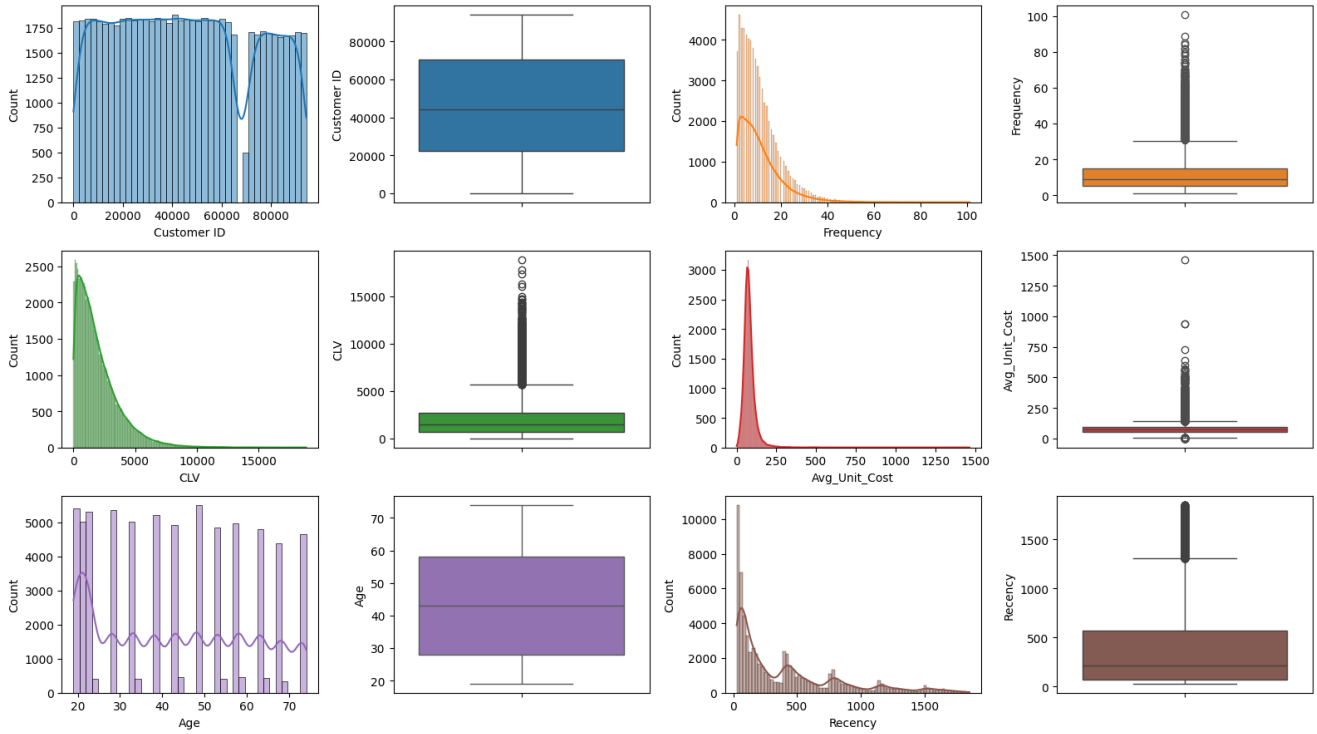
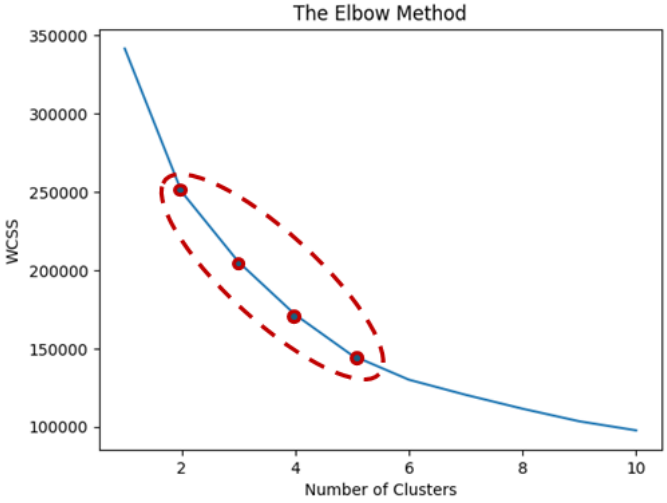


Figure 3 – Data Analysis Visualisation

Outliers were observed in all features apart from age and were retained as they may reflect genuine customer behaviour relevant to segmentation. Their validity has not been independently confirmed. Most feature distributions are non-normal, and extreme values may influence the clustering results.

4.2 Cluster Evaluation and Validation



WCSS - Elbow Plot Results show a soft elbow.

There is no sharp 'knee', we see:

- A very steep drop from k=1 -> 2
- A continued drop at k=3
- k ≥ 4, the curve flattens

We cannot determine a definitive value of k from this plot alone.

Figure 4 – Data Analysis Visualisation

Silhouette scores are similar across the tested values of k, with k = 5 achieving the highest average score (0.2655). Three clusters were selected as a practical balance between the diagnostics and business interpretability, rather than a clear statistical optimum.

Silhouette plots show clustering patterns and provide more actionable information. Plots with 4 and 5 clusters show significant variability in cluster size and higher occurrence of negative coefficients when compared to clusters of 2 and 3.

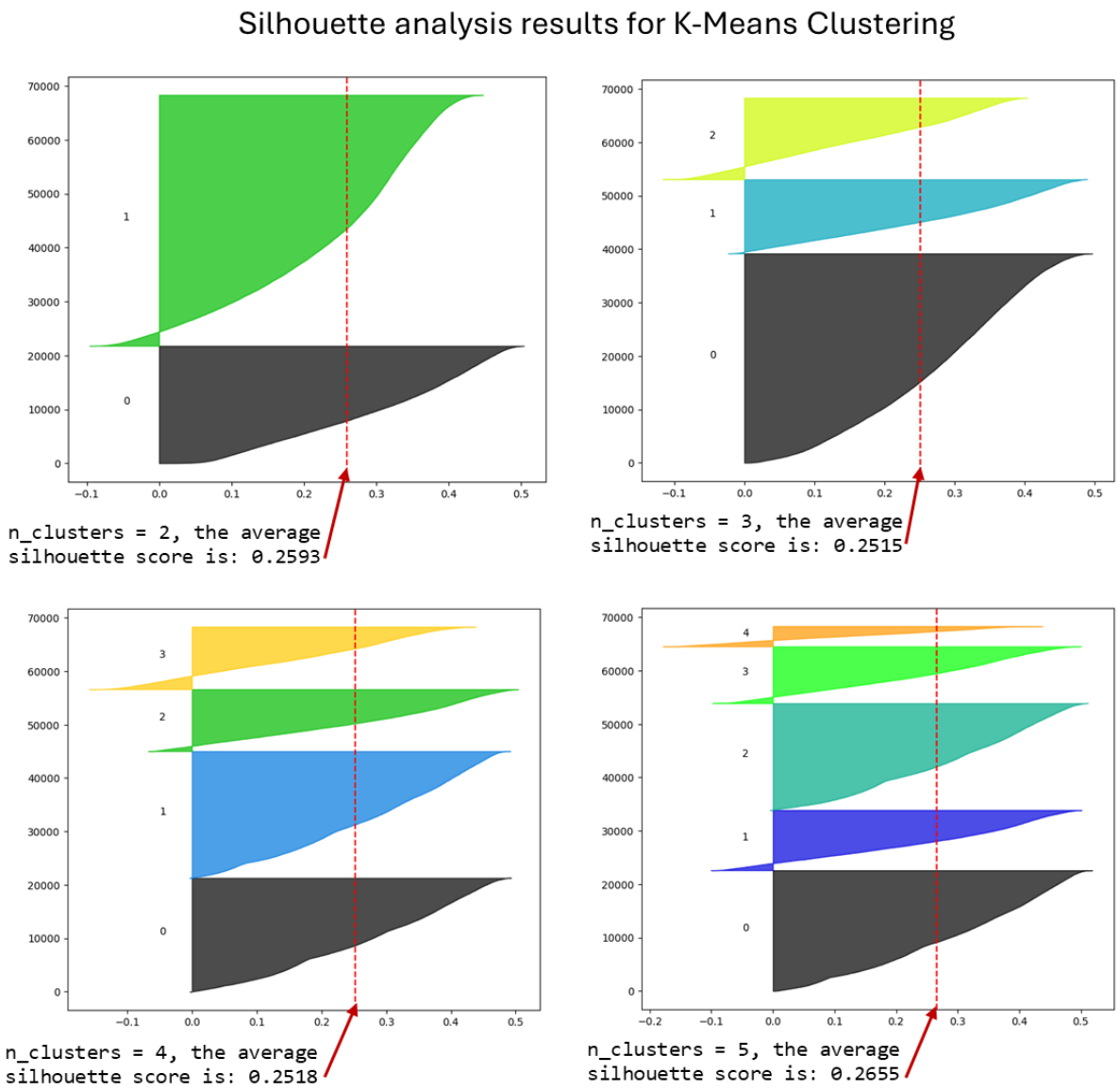


Figure 5 – Silhouette scores and plots

Dendrogram

Ward linkage minimises the increase in within-cluster variance at each merge. It remains sensitive to outliers. K = 2 represents the strongest split, and there are three well-formed groups with a relatively large distance gap between two and three.

A cut point at a distance of 290 produces three clusters, supporting k = 3 as a candidate.

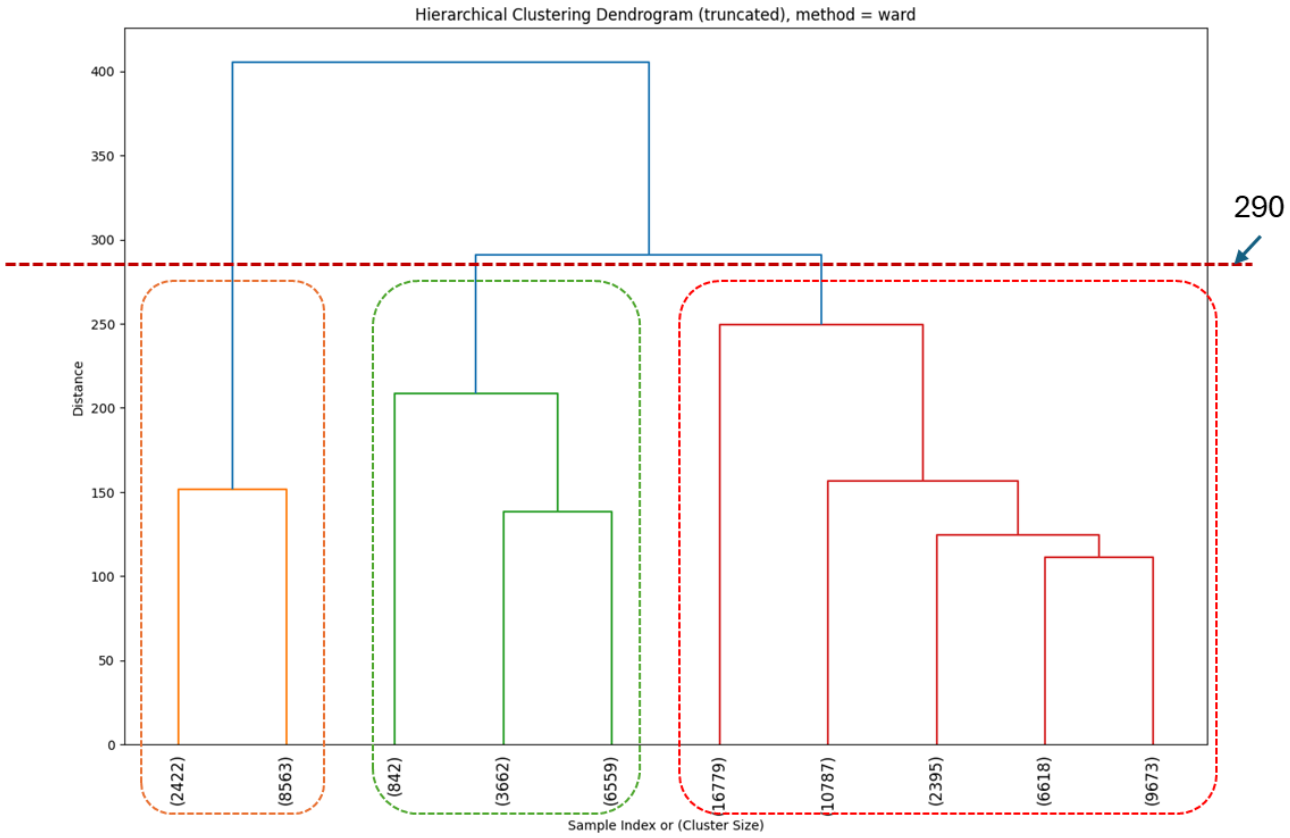


Figure 6 – Dendrogram (Ward's linkage method)

4.3 Selection of Number of Clusters

The Elbow method and silhouette scores identified a range of candidate values for k, but no clear optimal value.

The hierarchical dendrogram plot provided evidence of a clear division into three clusters, while the distance gap to 4 or more clusters was smaller and less distinct. Silhouette scores were similar across the candidate values, with k = 5 scoring highest.

On balance, three clusters is well supported by the analysis, and provides a practical compromise between adequate segmentation and complexity.

4.4 Customer Segment Profiles

NAME / CLUSTER	KEY CHARACTERISTICS	COMMERCIAL FOCUS
CULTIVATE CLUSTER 0 – (ACTIVE / MODERATE VALUE) BLUE	Moderate purchase frequency and CLV; moderate recency; more variable, slightly lower median unit cost; age overlaps dormant group Active but less consistent purchasing behaviour.	**Value growth and cross-sell.** Targeted promotions, personalised offers, and incentives to increase purchase frequency and basket size.
WIN-BACK CLUSTER 2 – (DORMANT / CHURNED) GREEN	Low purchase frequency and CLV; very high recency; higher median age; Largely inactive customers;	**Reactivation and churn prevention.** Targeted re-engagement campaigns, reminders, and selective incentives to drive re-engagement.
RETAIN CLUSTER 1 – RETAIN (CORE / LOYAL) ORANGE	Highest purchase frequency and CLV; lowest recency; moderate average unit cost; Consistent, high-value, frequently engaged customers.	**Retention and loyalty.** Loyalty rewards, exclusive offers, and upsell strategies to maintain engagement and lifetime value.

Table 1 - Customer Segmentation

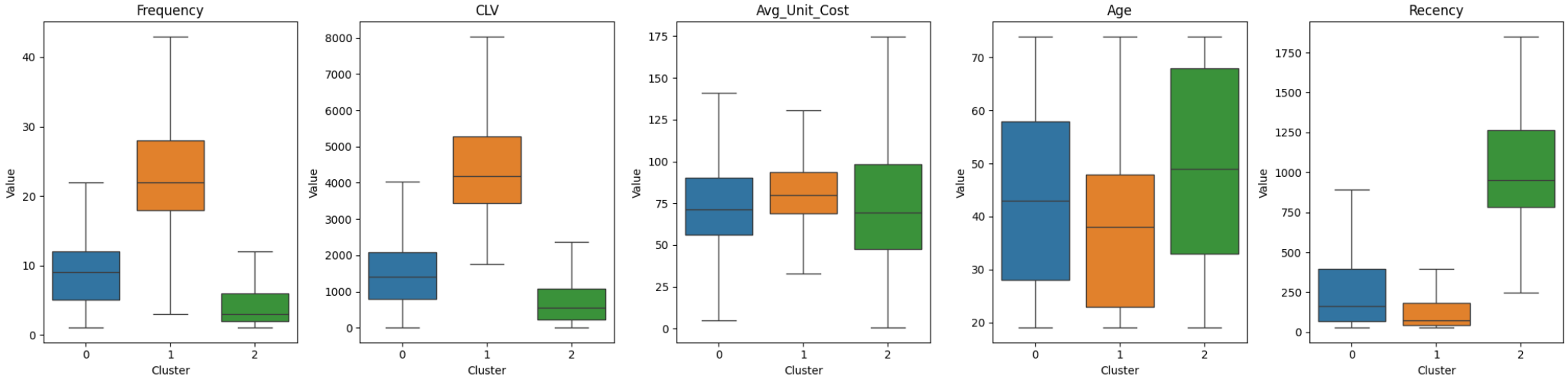


Figure 7 – Statistical Analysis of Cluster Features

For clarity of interpretation, box plots are shown with extreme outliers suppressed. All clustering and analysis were performed on the full dataset.

Three clusters are shown in a 2D scatter plot using PCA-reduced data, coloured to match the previous box plots. A t-SNE plot provides an alternative view of local relationships.

Clusters 0 and 2 display substantial overlap in these projections, while cluster 1 forms a more compact region with less overlap. K-means assigns each customer to one cluster; projected overlap does not imply soft clustering.

These plots support visual exploration but do not independently validate cluster separability. Feature-level similarities and differences are assessed through the original customer features and box plots.

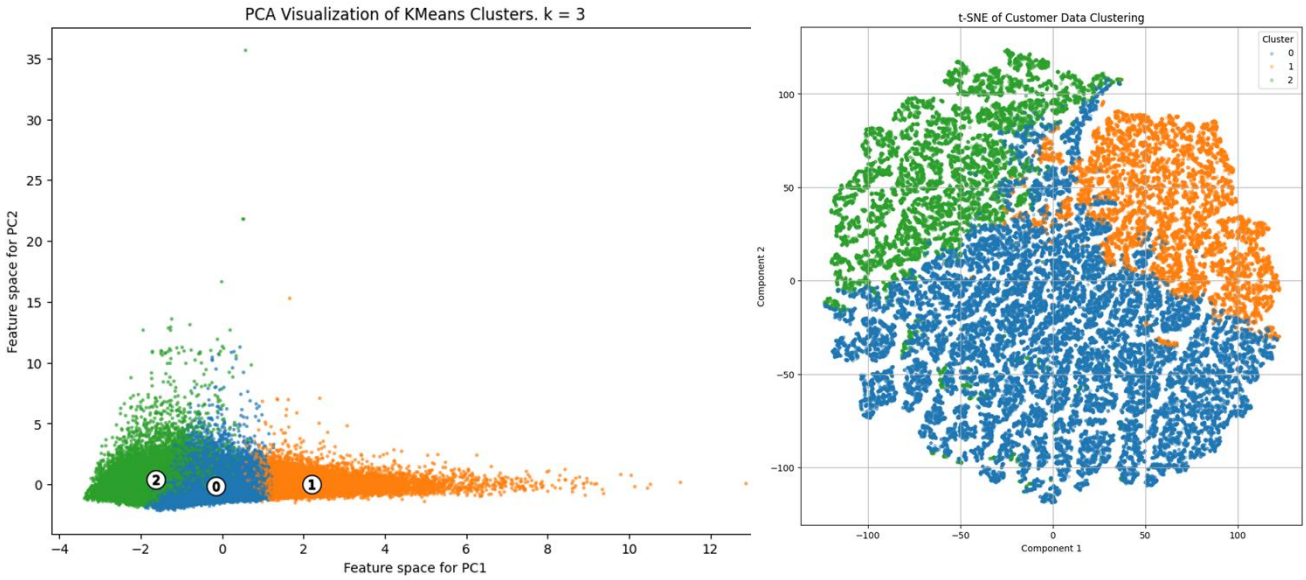


Figure 8 – PCA and t-SNE Visualisation of Clusters

5 Conclusions & Next Steps

Clustering and customer segmentation were successful in identifying distinct customer groups within the dataset. The selected number of clusters avoided overly broad groupings and suggested potential marketing actions requiring validation for each segment based on their underlying customer characteristics.

Next Steps

- Validate the identified customer segments and proposed marketing actions with relevant business stakeholders.
- Trial targeted marketing initiatives aligned to each segment and monitor performance metrics to assess effectiveness.

The lower-activity, longer-recency segment comprises 22.34% of customers. Subject to business validation and available marketing resources, a recency-based cut-off could be explored to separate inactive customers, followed by re-segmentation of the active customer base. Additional product-level metrics, such as brand preferences, could provide further insight into customer behaviour.

6 References

1. SAS, 2024. CUSTOMERS_CLEAN [Data set]. SAS. Last revised on 15 December 2021.